

第十二章 流体运动学

§ 12.1

1. 描述流动:

$$\left\{ \begin{array}{l} \text{拉格朗日 } \vec{v}(\vec{r}_0, t) \quad \text{质点的始位置} \\ \text{欧拉 } \vec{v}(\vec{r}, t) \quad \text{速度场} \end{array} \right.$$

2. 迹线、流线

迹线: 各质点的运动轨迹

$$\frac{d\vec{r}}{dt} = \vec{v}(\vec{r}_0, t)$$

流线: $d\vec{r} \parallel \vec{v}(\vec{r}, t)$

$$\frac{dx}{v_x(x, y, z, t)} = \frac{dy}{v_y(x, y, z, t)} = \frac{dz}{v_z(x, y, z, t)}$$

例: 已知速度场 $\vec{v} = cy\vec{i}$, 试求流线形状

$$\frac{dx}{cy} = \frac{dy}{0} = \frac{dz}{0}$$

$\rightarrow y = \text{const.}, z = \text{const.}$ 为直线族

例: 已知速度场 $\vec{v} = -\frac{cy}{x^2+y^2}\vec{i} + \frac{cx}{x^2+y^2}\vec{j}$, 试求流线形状

$$\frac{dx}{-\frac{cy}{x^2+y^2}} = \frac{dy}{\frac{cx}{x^2+y^2}} = \frac{dz}{0}$$

$$\rightarrow \frac{dy}{dx} = -\frac{x}{y}, \quad z = \text{const.}$$

即 $x^2 + y^2 = \text{const.}, z = \text{const.}$ 为圆族

流域 $\rightarrow$ 流量

$$\text{流量 } Q = \frac{dV}{dt} = \rho v S$$

$$\hookrightarrow \text{连续性方程: } \rho_1 v_1 S_1 = \rho_2 v_2 S_2 \quad (Q = \text{const.})$$

3. 当地变化率和实质变化率

$$\vec{a}(\vec{r}, t) = \frac{d\vec{v}}{dt}(\vec{r}, t)$$

$$\text{而 } \frac{d\vec{v}}{dt} \neq \frac{\partial \vec{v}(\vec{r}, t)}{\partial t}, \text{ 因速度还依赖于 } \vec{r}$$

$$\therefore \frac{d\vec{v}}{dt} = \frac{\partial \vec{v}(\vec{r}, t)}{\partial x} \dot{x} + \frac{\partial \vec{v}(\vec{r}, t)}{\partial y} \dot{y} + \frac{\partial \vec{v}(\vec{r}, t)}{\partial z} \dot{z} + \frac{\partial \vec{v}(\vec{r}, t)}{\partial t}$$

$$\text{实质变化率} = \frac{\partial \vec{v}}{\partial t} + \underbrace{(\vec{v}(\vec{r}, t) \cdot \nabla) \cdot \vec{v}}_{\text{当地变化率} \quad \text{漂移 (运动) 变化率}}$$

eg. 稳恒流动

$$\frac{\partial \vec{v}}{\partial t} = 0 \quad \frac{d\vec{v}}{dt} \neq 0$$

$$\Rightarrow \frac{d}{dt} = \frac{\partial}{\partial t} + \vec{v} \cdot \nabla$$

§ 12.2 速度场的分析

1. 场中点 O 径矢为 $\vec{r}$, 考察附近一点 P , 径矢为 $\vec{r} + d\vec{r}$

对 $\vec{v}(\vec{r})$ 做 Taylor 展开:

$$v_i(\vec{r} + d\vec{r}) = v_i(\vec{r}) + \sum_{\alpha=x,y,z} \left(\frac{\partial v_i}{\partial x_\alpha} \right)_{\vec{r}} dx_\alpha$$

$$\text{即 } dv_i = \sum_{\alpha=x,y,z} \left(\frac{\partial v_i}{\partial x_\alpha} \right)_{\vec{r}} dx_\alpha$$

$$\text{或 } \begin{pmatrix} dv_x \\ dv_y \\ dv_z \end{pmatrix} = \begin{pmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{pmatrix} \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix}$$

定义 张应变速率 $\rightarrow x$ 处在 $x+dx$ 处伸长速率为 $\frac{1}{2}$

$$e_{11} = \left(\frac{\partial v_x}{\partial x} \right)_r, e_{22} = \left(\frac{\partial v_y}{\partial y} \right)_r, e_{33} = \left(\frac{\partial v_z}{\partial z} \right)_r$$

切应变速率 $\rightarrow x$ 处在 $x+dx$ 处切应变速率为 $\frac{1}{2}$

$$= \left(\frac{\partial v_x}{\partial y} \right)_r$$

为了分离转动带来的切应变 (如 $\frac{1}{2} \frac{\partial v_z}{\partial x}$)

$$dV = dV_{\text{rot}} + dV_{\text{str}}$$

$$dV_{x\text{rot}} = e_{11} dx + \underbrace{\frac{1}{2} \left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right) dy}_{e_{12}} + \underbrace{\frac{1}{2} \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) dz}_{e_{13}}$$

$$dV_{y\text{rot}} = \underbrace{\frac{1}{2} \left(\frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y} \right) dx}_{e_{21}} + e_{22} dy + \underbrace{\frac{1}{2} \left(\frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y} \right) dz}_{e_{23}}$$

$$dV_{z\text{rot}} = \underbrace{\frac{1}{2} \left(\frac{\partial v_z}{\partial x} + \frac{\partial v_x}{\partial z} \right) dx}_{e_{31}} + \underbrace{\frac{1}{2} \left(\frac{\partial v_z}{\partial y} + \frac{\partial v_y}{\partial z} \right) dy}_{e_{32}} + e_{33} dz$$

$$dV_{x\text{str}} = \frac{1}{2} \left(\frac{\partial v_x}{\partial y} - \frac{\partial v_y}{\partial x} \right) dy + \frac{1}{2} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) dz$$

$$dV_{y\text{str}} = \frac{1}{2} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) dx + \frac{1}{2} \left(\frac{\partial v_y}{\partial z} - \frac{\partial v_z}{\partial y} \right) dz$$

$$dV_{z\text{str}} = \frac{1}{2} \left(\frac{\partial v_z}{\partial x} - \frac{\partial v_x}{\partial z} \right) dx + \frac{1}{2} \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) dy$$

$$\omega_x = \frac{1}{2} \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right)_r = \frac{1}{2} (\nabla \times \vec{v})_x$$

$$\omega_y = \frac{1}{2} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right)_r = \frac{1}{2} (\nabla \times \vec{v})_y$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right)_r = \frac{1}{2} (\nabla \times \vec{v})_z$$

$$\left. \begin{array}{l} \omega_x = \frac{1}{2} (\nabla \times \vec{v})_x \\ \omega_y = \frac{1}{2} (\nabla \times \vec{v})_y \\ \omega_z = \frac{1}{2} (\nabla \times \vec{v})_z \end{array} \right\} \vec{\omega} = \frac{1}{2} (\nabla \times \vec{v})$$

$\text{curl}(\vec{v})$

$$\Rightarrow d\vec{v}_{\text{rot}} = \vec{\omega} \times d\vec{r} \quad \vec{\omega} = \{\omega_i\}_{3 \times 1} \quad (\text{显然 } \vec{\omega} \text{ 是对称张量})$$

$$d\vec{v}_{\text{str}} = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix} d\vec{r} = \vec{\omega} \times d\vec{r}$$

$\Rightarrow$ P点运动 = 随O平动 + 纯切变 + 张应变 + 转动

考虑速度场的散度 $\underbrace{\vec{v} \cdot \vec{\nabla}}_{\text{div}(\vec{v})} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$

$$= e_{11} + e_{22} + e_{33}$$

$$= 0$$

其中 θ 为膨胀系数

$\Rightarrow$ 不可压缩流体: $\vec{v} \cdot \vec{\nabla} = 0$

例: 分析速度场

1) $\vec{v} = cy \vec{i}$

$$e_{11} = \frac{\partial v_x}{\partial x} = 0, \quad e_{22} = e_{33} = 0$$

$$e_{12} = e_{21} = \frac{1}{2} c, \quad e_{13} = e_{31} = e_{23} = e_{32} = 0$$

$$\omega_x = \frac{1}{2} (\vec{v} \times \vec{\nabla})_x = \frac{1}{2} \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) = 0$$

$$\omega_y = 0$$

$$\omega_z = \frac{1}{2} (0 - c) = -\frac{1}{2} c$$

$$dV_{x\vec{v}} = \frac{1}{2} c dy$$

$$dV_{x\vec{v}} = \frac{1}{2} c dy$$

$$dV_{y\vec{v}} = \frac{1}{2} c dx$$

$$dV_{y\vec{v}} = -\frac{1}{2} c dx$$

$$dV_{z\vec{v}} = 0$$

$$dV_{z\vec{v}} = 0$$

$$\vec{\omega} = \sum_i \omega_i \vec{e}_i = 0$$

$$a) \quad \vec{v} = \left[-\frac{cy}{x^2+y^2} \hat{i} + \frac{cx}{x^2+y^2} \hat{j} \right]$$

$$e_{i1} = \frac{\partial v_x}{\partial x} = -\frac{2cyx}{(x^2+y^2)^2} \quad e_{i2} = \frac{2cyx}{(x^2+y^2)^2} \quad e_{31} = 0$$

$$e_{i2} = e_{21} = \frac{1}{2} \left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right) = \frac{1}{2} \left[-c \frac{x^2+y^2-2y^2}{(x^2+y^2)^2} + c \frac{x^2+y^2-4x^2}{(x^2+y^2)^2} \right]$$

$$= \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$e_{13} = e_{31} = e_{23} = e_{32} = 0$$

$$\omega_x = \frac{1}{2} (\nabla \times \vec{v})_x = 0$$

$$\omega_y = 0$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) = 0$$

$$dV_{xz} = \frac{-2cyx dx + (y^2-x^2) dy}{(x^2+y^2)^2}$$

$$dV_{xz} = 0$$

$$dV_{yz} = \frac{(y^2-x^2) dx + 2cyx dy}{(x^2+y^2)^2}$$

$$dV_{yz} = 0$$

$$dV_{zx} = 0$$

$$dV_{zx} = 0$$

2. 有旋流动和无旋流动

$$(\vec{\omega} \neq 0) \quad (\vec{\omega} = 0)$$

如何判断? 小窍门

定义: 环流量 $\Gamma = \oint_C \vec{v} \cdot d\vec{l}$ 其中 C 为闭合曲线

↓ Stokes 公式

$$\Gamma = \oint_S \nabla \times \vec{v} \cdot d\vec{S} \quad \text{若 } S \text{ 较小, 近似为平面}$$

$$= \iint_S 2\vec{\omega} \cdot d\vec{S} \quad \Gamma = 2\vec{\omega} \cdot \vec{S}$$

例: 说明 $\vec{v} = -\frac{cy}{x^2+y^2} \vec{i} + \frac{cx}{x^2+y^2} \vec{j}$ 在原点有旋

取以原点为中心, 半径为 ϵ 的闭合曲线 $C_\epsilon: x^2+y^2 = \epsilon^2$

$$\text{法向: } \vec{n} = \frac{\vec{r}}{r} = \frac{x\vec{i} + y\vec{j}}{\sqrt{x^2+y^2}} = \frac{x}{\epsilon} \vec{i} + \frac{y}{\epsilon} \vec{j}$$

$$\text{切向: } \vec{t} = \frac{-y\vec{i} + x\vec{j}}{\sqrt{x^2+y^2}} = -\frac{y}{\epsilon} \vec{i} + \frac{x}{\epsilon} \vec{j}$$

$$\Gamma_\epsilon = \oint_{C_\epsilon} \vec{v} \cdot d\vec{l} = \oint_{C_\epsilon} \vec{v} \cdot \vec{t} d\ell = \vec{v} \cdot \vec{t} \oint_{C_\epsilon} d\ell$$

$$= \frac{cy^2 + cx^2}{(x^2+y^2)\epsilon} \cdot 2\pi\epsilon = 2\pi c \neq 0$$

$$\omega = \frac{\Gamma}{2S} = \frac{c}{\epsilon} \rightarrow \infty \quad \text{在不是“足够小”的 } C_\epsilon \text{ 上又如何?}$$

启示: 引 ω 一些曲线, 使其与当地 $\vec{\omega}$ 相同

$$\text{div}(\vec{\omega}) = \nabla \cdot \vec{\omega} = \nabla \cdot \left(\frac{1}{2} \nabla \times \vec{v} \right) = \frac{1}{2} \nabla \cdot \nabla \times \vec{v}$$